WESAD Emotion Recognition: Comprehensive 10–15 Page Expanded Report

(With placeholders for figures, tables, and diagrams)

A. METHODOLOGY AND PREPROCESSING

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1. Introduction

The WESAD (Wearable Stress and Affect Detection) dataset is one of the most influential physiological emotion-recognition datasets available for research. Its strength lies in its multimodal sensor inputs, realistic recording procedures, and well-defined emotional elicitation protocol. This study aims to construct an end-to-end, fully reproducible pipeline capable of recognizing four emotional states—baseline, stress, amusement, and meditation—from these physiological signals. The approach integrates careful preprocessing, feature-level synchronization, a hybrid deep learning architecture, and extensive Leave-One-Subject-Out (LOSO) evaluation.

[PLACEHOLDER: FIGURE – WESAD sensor setup diagram]

2. Dataset Overview

WESAD contains recordings from 15 subjects (S2–S17). Each participant wears a chest-mounted RespiBAN device and a wrist-mounted Empatica E4 device. The signals captured include:

• Chest (~700 Hz): ECG, respiration, EDA, temperature, 3-axis accelerometer

• Wrist (~32 Hz): EDA, temperature, 3-axis accelerometer

Each subject participates in controlled emotional states generated through validated psychological methods: a calm baseline, a stress induction phase (e.g., Trier Social Stress Test), an amusement segment (humor videos), and a meditation phase.

3. Motivation for LOSO Cross-Validation

Emotion signatures vary drastically across individuals. Rather than train and test on mixed-subject data—which inflates accuracy due to subject-specific biases—LOSO tests the model’s ability to generalize to unseen users. This makes the evaluation realistic for wearable deployment.

[PLACEHOLDER: TABLE – LOSO fold definitions and subject mapping]

4. Signal Processing Pipeline

The raw sensor streams differ in sampling rates, units, and signal noise characteristics. To build a coherent multimodal feature tensor, the following pipeline is applied:

4.1 Acceleration Magnitude

Tri-axial accelerometer signals are condensed into a single motion magnitude per device to reduce noise and redundancy.

4.2 Resampling for Temporal Alignment

The chest signals are downsampled by a factor of ~21 to approach the wrist sampling rate. Meanwhile, wrist signals are interpolated upward to match the downsampled chest signal lengths. This ensures aligned temporal grids across all eight final channels.

4.3 Window Extraction

The continuous streams are segmented into overlapping 10-second windows. A 50% overlap increases the quantity of training samples without introducing information leakage.

4.4 Label Assignment

Because labels exist at the second-level timescale, each window receives the majority class present within its duration.

4.5 Normalization

Z-scoring is computed per channel using only the training subjects' statistics to avoid leakage. Normalization helps stabilize gradient flow and reduces inter-subject physiological variability.

4.6 Data Distribution and Class Imbalance

The distribution is heavily skewed: baseline comprises ~68% of samples, while amusement accounts for only ~7.5%. This imbalance requires corrective strategies such as weighted loss.

[PLACEHOLDER: FIGURE – class distribution histogram]

B. MODEL ARCHITECTURE AND LEARNING PRINCIPLES

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1. Design Philosophy

Multimodal physiological sequences contain both local patterns (short-term oscillations, sharp peaks) and long-range temporal dependencies (slow respiration trends, tonic EDA changes). Therefore, the architecture integrates three complementary components:

- Convolutional layers for local temporal feature extraction
- Bidirectional GRU layers for sequential modeling
- Multi-head attention for adaptive temporal weighting

2. CNN Frontend

Two 1D convolutional layers first transform the raw input (8 channels) into deeper temporal feature maps (64 channels). These layers automatically detect patterns such as heart-rate fluctuations or respiratory waves.

[PLACEHOLDER: FIGURE – CNN filter visualization]

3. Bidirectional GRU

The recurrent layers capture long-range temporal structure in both forward and backward directions. Bidirectionality improves feature richness by making each timestep aware of both past and future signals within a window.

4. Layer Normalization

Applied after recurrent layers, LayerNorm stabilizes the distribution of activations and improves convergence.

5. Multi-Head Self-Attention

Attention allows the network to focus on informative timepoints such as stress-induced EDA spikes or amusement-driven respiration modulation. With two attention heads, the model learns complementary relevance patterns.

[PLACEHOLDER: FIGURE – attention heatmap sample]

6. Temporal Pooling

Mean pooling summarizes a sequence into a fixed-dimensional representation while preserving its overall temporal characteristics. It allows efficient downstream classification.

7. Classification Head

A two-layer fully connected network (256 → 128 → 4) performs emotion classification. The output is a four-class probability distribution.

8. Training Strategies

The model uses AdamW optimization with a learning rate of 1e-3, ReduceLROnPlateau scheduling, and weighted cross-entropy to counter class imbalance. Gradient clipping (norm 5.0) prevents exploding gradients during recurrent operations. Automatic Mixed Precision is used to accelerate training on GPU.

[PLACEHOLDER: TABLE – hyperparameters summary]

9. Initialization Principles

Kaiming initialization supports ReLU-based convolution layers, while Xavier initialization ensures stable variance propagation in recurrent units.

C. RESULTS INTERPRETATION AND DISCUSSION

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1. Overall Accuracy and Limitations

The model achieves a mean accuracy of 63.1%. However, due to heavy class imbalance, accuracy is misleading as a standalone indicator. Many models can achieve >60% simply by predicting baseline frequently.

2. Macro F1-Score

A more reliable metric for imbalanced datasets, macro F1 averages recall and precision across classes. The observed macro F1 of ~30.33% reflects difficulty in minority-class detection.

[PLACEHOLDER: TABLE – per-class precision/recall/F1]

3. Per-Class Performance

• Baseline (0.75 F1): Strong performance because it dominates the dataset and exhibits stable physiological patterns.

• Meditation (0.35 F1): Distinct respiration and low-arousal EDA help differentiation.

• Stress (0.25 F1): Often confused with baseline due to overlapping physiological responses.

• Amusement (0.10 F1): Most challenging because short bursts of laughter may resemble stress or baseline depending on the individual.

4. Confusion Matrix Interpretation

Confusion matrices reveal systematic misclassification patterns:

• Baseline is over-predicted.

• Stress is confused with baseline due to similar arousal levels.

• Amusement overlaps with stress in EDA/ACC profiles.

• Meditation is moderately distinguishable but still subject-dependent.

[PLACEHOLDER: FIGURE – confusion matrix heatmap]

5. Subject-Wise LOSO Variability

Macro F1 ranges from 14.8% (S8) to 57.6% (S9). Some subjects display clean, well-separated physiological responses, while others introduce noise, motion artifacts, or atypical stress reactions.

6. Role of Imbalance

The extreme imbalance (approx. 9:1) between baseline and amusement yields poor minority detection. Weighted loss partially compensates, but structural improvements (augmentation, focal loss, generative oversampling) are recommended.

7. Observed Overfitting

Training converges by epochs 15–25. Slight overfitting is visible from divergence between validation and training curves, especially in minority-class recall.

8. Computation and Deployment

Training is efficient (~6.5 s/epoch). Inference requires only 10–20 ms per window and the model occupies ~3 MB, making it suitable for wearable deployment.

FUTURE DIRECTIONS

• Data augmentation (e.g., jittering, frequency warping) for diversity

• Focal loss to address extreme imbalance

• Early-fusion vs. late-fusion multimodal experiments

• Self-supervised pretraining on raw physiological streams

• Transformer-based architectures for improved temporal modeling

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END OF REPORT
